

Highlights

Decision before Feature: Deciding Whether to Execute a Landscape Descriptor Query in Black-box Optimization

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- Formulates landscape-descriptor acquisition as a decision made before descriptor computation.
- Uses offline, state-aligned utility labels and algorithm-agnostic search behavior from native optimizer histories.
- Specifies train-only model selection and threshold fitting, with development and prospective external evaluations explicitly separated.

Decision before Feature: Deciding Whether to Execute a Landscape Descriptor Query in Black-box Optimization

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ABSTRACT

Landscape descriptors are widely used to support algorithm selection in black-box optimization, but acquiring them consumes function evaluations, runtime, and memory, while their value may vary across search states. This paper formulates acquisition of a fixed landscape-descriptor query as a decision that precedes feature computation. The proposed Decision-before-Feature protocol reuses algorithm-agnostic, permutation-invariant behavior from complete native optimizer-update histories, constructs state-aligned joint, Behavior-only, and operational-increment labels through paired offline state-to-terminal future paths, and freezes a supervised decision rule under fully nested function out-of-fold evaluation. The primary query is a predefined 14-dimensional low-cost descriptor configuration; two larger configurations are reserved for representation-and-cost robustness analyses. **TBD** The prevalence and interval for non-beneficial primary-query states remain undetermined. **TBD** The nested-OOF model comparison and B3-versus-T0 increment remain undetermined; BBOB validation is a previously inspected internal development set rather than an independent confirmation set. **TBD** The end-to-end comparison with the prespecified baselines, including query use, final performance, and resource measures, remains undetermined. **TBD** Performance on the inspected BBOB-validation and CEC2017 development sets and on the prospective CEC2022 and engineering evaluations remains undetermined; no transfer claim is made. **TBD** The association between search behavior, search maturity, and query utility remains undetermined and has a descriptive, non-causal interpretation.

1. Introduction

Black-box optimization is used when objective values can be obtained by evaluation but an analytic model, derivative information, or a tractable problem structure is unavailable. In this setting, the choice of optimizer matters: algorithms expose complementary behavior across problem instances, while no member of a practical portfolio is uniformly preferable. Automated algorithm selection addresses this heterogeneity by mapping an instance representation to a portfolio decision. A common workflow evaluates a space-filling design, computes exploratory landscape analysis (ELA) features, and then applies a supervised selector [? ?]. This workflow makes landscape analysis a prerequisite for selection.

That prerequisite is not innocuous. Landscape characterization comprises families of measures with different assumptions, sampling requirements, search dependence, and computational demands [? ?]. The objective evaluations used to construct a landscape sample consume budget that could otherwise be assigned to optimization, and the appropriate feature-computation fraction depends on the selection scenario [?]. Contemporary systems therefore count feature evaluations or omit expensive groups under a restricted budget [?]. An ELA-informed portfolio decision may be useful and still fail to justify the resources spent to obtain its descriptors at a particular search state.

At the same time, an optimization prefix already produces information. Search-trajectory networks characterize convergence patterns, recurrent transitions, and attraction

regions from ordinary optimizer runs [?]. DynamoRep preserves the longitudinal evolution of population summaries, while Opt2Vec learns from populations visited during search [? ?]. Probing trajectories use short objective-value sequences for a subsequent portfolio choice, and later work shows that their evaluation depends on both the classifier and the problem-level split [? ?]. Reusing evaluated points avoids a separate objective-function sample, but representation and inference still have computational cost. More importantly, problem classification or direct algorithm selection does not establish that a pre-query behavior summary predicts the net value of acquiring an independent descriptor query.

Generalization is a second obstacle. Algorithm-selection performance can appear strong when training and evaluation sets contain closely related instances. Under demanding problem and problem-combination splits, several classical and learned landscape representations did not improve over a single-best-solver baseline in the evaluated affine-BBOB setting [?]. The closest per-run warm-starting study likewise reported that results obtained on BBOB did not transfer directly to YABBOB [?]. Recent reviews identify heavy reliance on BBOB, heterogeneous feature protocols, and limited cross-benchmark evidence as continuing concerns [?]. Random instance partitions and within-training scores are therefore insufficient evidence for transfer to unseen function families or benchmark suites.

This paper studies a decision that precedes feature acquisition: given an optimization prefix and its algorithm-agnostic behavioral history, should a fixed landscape-descriptor query be executed at the current state? We call this

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setting *Decision-before-Feature*. It differs from trajectory-based per-run selection, which constructs a trajectory representation and chooses a second-stage algorithm [? ?], and from dynamic algorithm selection, which repeatedly changes search actions during a run [?]. It also differs from landscape-aware operator or population control [? ?]. The object here is not a parameter, operator, or optimizer update rule, but whether to acquire additional information before a downstream portfolio decision.

We formulate query value as the difference in clipped terminal $\log_{10}$ gap between a native no-query continuation and a query-plus-selector path from the same complete optimizer state, less a weighted $\log_{10}$ ratio of their state-to-terminal future-path runtimes. The primary time for each path is the median of three prespecified censored replays: a completed replay retains its observed time, whereas a timed-out or failed replay contributes the larger of its observed time and role-specific timeout. The already executed shared prefix is a sunk cost; full FE-zero-to-terminal wall-clock remains a separate policy endpoint. Memory is measured and reported separately in the primary analysis, for which $\lambda_M = 0$. The query evaluations already reduce the remaining optimization budget and are not deducted twice. Query-sample evaluations do not enter the optimizer population, but their best point and observed target hit enter the operational Query endpoint. We distinguish an observed target hit, path completion, and completion-conditioned endpoint success; standard ERT uses the observed hit even when a later path failure occurs. Offline labels use fold-specific downstream selectors, whereas the deployed gate observes only behavior available before the query. Query descriptors, function identity, algorithm identity, optimizer-specific parameters, and known-optimum gaps are excluded from that gate. The resulting question is not whether landscape features can sometimes support selection, but when the expected downstream difference of the evaluated query is sufficient to justify its acquisition cost.

The study is organized around five research questions. RQ1 characterizes the statewise utility of the primary descriptor_cheap_invariant query. RQ2 tests whether algorithm-agnostic behavior adds decision value beyond elapsed budget and a dimension-stratified time-only comparator. RQ3 compares the resulting policy with fixed, random, traditional landscape-based, behavior-only, and portfolio references under common resource accounting. RQ4 reports the inspected six-function BBOB-validation set and CEC2017 as development evaluations, and reserves CEC2022 and engineering problems for prospective evaluation after their complete configurations and endpoints have been fixed. Each suite is reported separately. The BBOB-validation estimand is a finite-set mean, not a function-superpopulation or independent-confirmation estimand. RQ5 examines the six prespecified T0/B1/B2/B2+Motion/B2+Maturity/B3 groups, model-appropriate coefficient or discriminant stability, maturity-utility associations, and variation across function, dimension, and query configuration. RQ4 and

RQ5 remain empirical questions; their intended evaluations are not evidence that transfer or explanatory stability has already been obtained.

The contributions are methodological. First, we define execution of a fixed landscape-descriptor query as a resource-aware prediction problem with a state-conditional net-utility target. Second, we construct the gate input from permutation-invariant native-update histories while excluding algorithm identity and query descriptors. Third, we separate this gate from a query-specific downstream action-loss selector and make native continuation and population-transfer initialization explicit. Fourth, we specify nested grouped-by-function out-of-fold model selection and train-only threshold fitting, together with an evaluation structure connecting decision utility, final optimization loss, query use, and measured resource cost. The grouping unit is function ID, not a claim of generalization across a classical landscape-family taxonomy. Among the literature reviewed here, we did not identify the same combination of pre-query information timing, matched-state utility labels, and an independently acquired query. Whether the method yields a predictive or resource advantage is determined only by RQ1–RQ5.

The remainder of the paper reviews landscape-based and trajectory-based selection, behavior characterization, adaptive optimization, and generalization; formalizes the decision and utility target; describes the offline construction and learning procedure; and presents the RQ-aligned evaluation and reproducibility design.

2. Related work

2.1. Landscape representations and automated algorithm selection

Fitness-landscape analysis describes properties of an optimization problem that may be relevant to search, including ruggedness, modality, neutrality, fitness distributions, and variable interactions. The corresponding measures differ in their assumptions, sampling procedures, dependence on a search process, and computational demands [?]. ELA operationalizes this idea through numerical descriptors computed from sampled objective values. Its original formulation proposed relatively inexpensive, computer-generated features as a route from problem characterization to automatic algorithm selection [?]. Software such as pflacco now exposes multiple feature families with different sample requirements and domains of applicability [?]. Estimated ELA values can also depend materially on the sampling strategy, and a mismatch between the designs used for training and deployment can degrade the resulting predictor [?]. The sample design is therefore part of the representation, not a neutral implementation detail.

In automated algorithm selection (AAS), a conventional pipeline computes an instance representation and predicts which portfolio member should be used. ELA combined with supervised learning has been evaluated against the single best solver (SBS) and virtual best solver (VBS), often

under leave-one-function-out or related problem-oriented splits [?]. More recent systems retain the sample–feature–selector structure while changing the portfolio, feature set, or learning model [?]. These studies establish the downstream use of landscape information. They do not determine whether an independently sampled descriptor query should be acquired at a particular state before its features exist.

2.2. Feature acquisition as a resource cost

Landscape information is acquired rather than supplied. Objective evaluations used for feature sampling reduce the budget left for optimization, while feature computation adds computational demand. Reusing the sample as an optimizer population can reduce the effective FE charge in a particular pipeline [?], but this does not make the representation cost-free in general. Runtime and peak memory are therefore measured separately in the present study. A recent preprint study of feature budgets found that the preferred sampling fraction varied with the portfolio, problem set, dimension, target budget, and selection scenario [?]. Similarly, ELA-based selection experiments may restrict feature groups when the acquisition budget is limited [?].

These findings motivate cost-aware evaluation, but their decision variable is usually the portfolio action after features have been obtained. In contrast, Decision-before-Feature compares executing and skipping the same prespecified query from a shared search state. Query FEs enter through the reduced continuation budget rather than a second penalty; normalized time is penalized explicitly, while memory is reported and examined only in sensitivity analysis. The target is therefore not a classifier for whether a feature family is “cheap.”

2.3. Trajectory-based and per-run algorithm selection

The closest antecedents replace an independent landscape sample with information collected during an initial optimizer run. Jankovic et al. begin with CMA-ES, compute trajectory-based landscape features, predict the performance of candidate second-stage optimizers, and warm-start the selected one [?]. In that formulation, the trajectory representation is constructed for every run at a fixed switch point; the decision is which second-stage optimizer to use, not whether to construct the representation. Kostovska et al. extend this per-run setting to compare trajectory ELA, time series of CMA-ES internal state variables, and their combination [?]. The latter study also reports that its BBOB findings did not transfer directly to YABBOB, showing that such transfer cannot be assumed; trajectory-coverage mismatch is discussed there as a possible explanation. Both internal evaluations retain the same BBOB functions across training and test partitions: the CEC study groups five-dimensional observations by instance identifier, while the PPSN study uses run- and instance-level partitions in five and ten dimensions. These protocols answer different questions from the present grouped-by-function OOF design; they are antecedents for representation and warm-starting, not precedents for the present evaluation design.

Probing trajectories provide another algorithm-centric representation. A short run records an objective-value sequence, and the resulting time series supports a later portfolio choice [?]. A subsequent benchmark compared multiple time-series classifiers under both leave-one-instance-out and the harder leave-one-problem-out evaluation [?]. This result is relevant here because it shows that the predictive model and problem partition remain part of a trajectory selector’s empirical specification. Nevertheless, probing is performed before every selection, the target is the best portfolio algorithm, and the output is an optimizer choice. The probe itself still consumes evaluations even though it is not a separate landscape sample. It does not construct paired skip/query utility for a separate landscape query or decide whether that query should be acquired.

2.4. Trajectory representations and behavior characterization

Other work studies what optimizer histories represent rather than which action to take. Search trajectory networks summarize visited regions and transitions for comparative analysis and visualization [?]. DynamoRep concatenates generation-wise coordinate and fitness summaries, thereby preserving longitudinal population dynamics without a separate objective sample; Opt2Vec instead learns a representation from populations observed during a run [? ?]. DynamoRep fits separate classifiers for each optimizer and evaluates three-dimensional BBOB class identification, while Opt2Vec likewise reports problem classification. These tasks neither establish a fixed-dimensional, algorithm-agnostic state representation nor predict the utility of acquiring a separate landscape query.

Behavioral analysis provides a complementary vocabulary. Hayward and Engelbrecht characterize exploration, exploitation, locality, communication, and evaluation effort with a suite of whole-run measures, explicitly treating behavior as problem dependent [?]. Some of those measures rely on a known optimum, persistent individual identity, or algorithm interaction structure and hence are unsuitable as pre-query inputs here. Mbasso et al. use distance-to-reference decay, directional entropy, and stagnation to diagnose late-stage behavior and trigger optimizer-specific interventions [?]. Only the distance-decay and stagnation semantics motivate the present state representation: distance is redefined relative to the current population best, while particle-wise directional entropy is excluded because it requires persistent cross-generation identities. The reported interventions do not validate the study-specific Search Maturity construction or establish that behavior predicts landscape-query utility. More generally, computability alone does not make a behavior indicator informative or complementary: sensitivity, redundancy, single-run computability, and the information available at deployment all constrain its use [?].

2.5. Adaptive search and dynamic algorithm selection

Landscape-aware adaptive methods change the search mechanism after observing landscape or performance information. Examples include mutation-operator selection in differential evolution and adaptation of subpopulation number and partitioning in artificial bee colony optimization [? ?]. Adaptive landscape analysis instead studies how locally sampled landscape descriptors evolve during an optimizer run, with the longer-term aim of online selection or configuration [?]. It therefore computes descriptors on 2,000 additional points sampled from the current CMA-ES distribution at prescribed target levels. “Cheap” denotes that the chosen descriptor families need no further evaluations beyond that sample, not that acquisition is free. The preliminary study neither makes a pre-descriptor acquisition decision nor couples the descriptors to a paired skip/query utility. Dynamic algorithm selection can make repeated choices during a run; RL-DAS, for example, represents the process as a reinforcement-learning problem over differential-evolution variants using landscape and algorithm-history information [?]. Dynamic algorithm configuration makes the control problem explicit through a state space, an action space over hyperparameter settings or components, a reward, and a decision frequency; DACBench includes both CMA-ES step-size control and repeated selection among modular evolutionary-algorithm components [?]. Parameter control and MetaBBO cover still broader choices over parameters, components, structures, and learned update rules [? ?]. The broader adaptive-operator-selection literature likewise separates stateless credit-based rules from state-based policies that map solution, problem, or optimization-process features to an operator [?]. In both cases, the selected action is a search operator; the acquisition of an independent descriptor query is not the action being evaluated.

Those approaches optimize a search action or solver mechanism. The present method uses offline supervised learning and leaves the portfolio algorithms unchanged. Its gate decides whether to acquire information; only after a call does a separate, fixed action-loss selector choose between native continuation and population-transfer initialization. Thus, its action is neither an optimizer hyperparameter nor a repeated component choice. Table 1 summarizes this distinction for the closest task formulations.

2.6. Evaluation splits and the remaining gap

Representation quality and selector performance depend strongly on how problems are divided between training and evaluation. In an affine-BBOB study, several landscape representations that performed well under easier splits did not outperform SBS under the most demanding problem-oriented splits [?]. A recent representation survey likewise highlights inconsistent sampling and split protocols, heavy dependence on BBOB, and limited cross-benchmark evidence [?]. These observations motivate function-ID-grouped out-of-fold selection and function-held-out evaluation; they do not guarantee that the resulting gate transfers.

In the present study, the BBOB-validation outcomes were inspected during earlier development, so that split supplies an internal finite-set estimate rather than an independent confirmation. Benchmarking work further shows that leave-one-instance-out evaluation can reward representations that recover a problem class shared by related training and test instances without learning to rank algorithms on unseen problems, and that scale-sensitive prediction errors need not measure selection quality [?]. The present study therefore uses function-ID-grouped partitions and selects the Decision Model by policy utility rather than raw action-loss regression error. This design addresses those two failure modes without establishing generalization in advance.

Among the literature reviewed here, we did not identify a method that makes the same decision at the same information time: before descriptors from a prespecified independent query exist, use only algorithm-agnostic behavior to predict whether the query’s downstream performance difference justifies its FE and computational cost. Decision-before-Feature addresses this corpus-scoped gap with matched-state offline labels, an explicit separation between the query gate and the downstream selector, a time-only comparator, and nested nested function-level evaluation. The literature supports this formulation and its design constraints; it does not supply evidence for the pending RQ1–RQ5 outcomes or for generalization across function IDs or benchmark suites.

3. Problem formulation

3.1. State-conditional analysis selection

Let $p \in \mathcal{P}$ denote a bounded continuous black-box minimization problem,

$$\min_{x \in \mathcal{X}_p} f_p(x), \quad \mathcal{X}_p = [\ell_p, u_p] \subset \mathbb{R}^d, \quad (1)$$

evaluated under a fixed total budget $B = FE_{\text{total}}$. The portfolio is

$$\mathcal{A} = \{\text{DE, PSO, CMA-ES, SHADE}\}. \quad (2)$$

At decision opportunity t , the native optimizer state is

$$S_t = (X_t, y_t, H_t, \omega_t, e_t), \quad (3)$$

where X_t and y_t are the evaluated population and objective values, H_t is the retained native-update history, ω_t contains optimizer dynamics and random-number-generator state, and e_t is the exact number of consumed FEs. Every continuation starts from this complete state; a population-only reconstruction is not treated as native continuation.

The analysis-selection decision is

$$d_t = \begin{cases} 0, & \text{continue without acquiring query } q, \\ 1, & \text{acquire } q \text{ and invoke its selector.} \end{cases} \quad (4)$$

The primary query, `descriptor_cheap_invariant`, comprises 14 fixed, permutation-invariant, low-cost descriptors computed from an independent Latin-hypercube sample of size

Table 1

Closest task formulations compared by decision object, information time, acquisition cost, and state transition. “Extra FEs” denotes an independent landscape sample beyond an existing prefix or probe.

Approach	Decision target	Information at decision	Extra FEs	Frequency and transition
Static ELA-based AAS [? ?]	Portfolio algorithm	Acquired landscape descriptors	Usually yes	Once, before the chosen optimizer starts
Jankovic et al. [?]	Second-stage algorithm	ELA computed from a fixed CMA-ES prefix	No separate sample	Once, with algorithm-specific warm-starting
Kostovska et al. [?]	Second-stage algorithm	Trajectory ELA and/or CMA-ES state series	No separate sample	Once, with algorithm-specific warm-starting
Probing trajectories [? ?]	Portfolio algorithm	Objective-value sequence from a short probe	No separate landscape sample	Once, after probing
RL-DAS [?]	Differential-evolution variant	Landscape, search history, and context	Not isolated as a fixed query	Repeated online scheduling
DACBench [?]	Hyperparameter or algorithm component	Target-algorithm state and instance context	Not a landscape-query decision	Repeated dynamic configuration
Decision-before-Feature	Execute or skip one fixed query	Pre-query algorithm-agnostic behavior	Only when called	Opportunities until one call; native continuation or population transfer

50d. It consumes $FE_q = 50d = 0.05B$ objective evaluations. Two response summaries that become constants after the prescribed median/IQR preprocessing are excluded from the active descriptor list. This removal changes neither the query identifier, sample design, nor action outcomes. Two larger, prespecified query configurations evaluate representation and acquisition-cost dependence. Thus, d_t does not denote arbitrary landscape analysis, and the primary result cannot be extended to unevaluated descriptor families.

3.2. Shared state, actions, and budgets

The default is the fold-specific single best solver (SBS), computed from complete-budget training outcomes by equally weighted aggregation over runs, problems (dimension–instance pairs), and functions of clipped $\log_{10}$ gap. In the primary population, the prefix and default algorithms are this same SBS. The Skip path therefore restores and continues the complete native state for

$$B_t^s = B - e_t \quad (5)$$

additional FEs.

At a state with prefix algorithm a_t , the non-duplicated action set is

$$\mathcal{A}(S_t) = \{\text{continue_current}\} \cup (\mathcal{A} \setminus \{a_t\}). \quad (6)$$

The current action preserves ω_t and advances natively. Each other action performs one population-transfer initialization from the evaluated population, its objective values, and the best-so-far point; algorithm-specific internal state is newly initialized. Query sample points are not inserted into the optimizer population.

The Query and Behavior-only action matrices use different continuation budgets,

$$B_t^q = B - e_t - FE_q, \quad (7)$$

$$B_t^b = B - e_t. \quad (8)$$

The four observed outcomes in either matrix supervise a multi-output action-loss regressor through the within-state transformation

$$\tilde{L}_a = \frac{L_a - \min_c L_c}{\max(\max_c L_c - \min_c L_c, 10^{-12})}. \quad (9)$$

The minimum over actions that were actually run is called the *best observed action*, not an oracle.

3.3. Terminal endpoints and decision-state future-path timing

One prespecified Stage-A scientific execution fixes each path’s terminal gap, observed first hit, target-hit status, path completion, completion-conditioned endpoint success, planned FE, effective FE, and failure status. An observed target hit is retained even if the path later fails; standard ERT uses observed target hit rather than completion-conditioned endpoint success. Let g_s and g_b be the Stage-A terminal best gaps of the Skip and Behavior-only paths. Although the query sample is not an optimizer population, every sampled point is a real objective evaluation. The Query terminal gap is therefore

$$g_q = \min\{g_{q,\text{sample}}, g_{q,\text{continuation}}\}. \quad (10)$$

The same best-so-far endpoint defines Query target hit and first hitting time. A hit during sampling records its within-query offset. The continuation-only gap and the sample-best contribution remain separate diagnostics.

Each raw gap is clipped according to the prespecified suite endpoints and then log transformed,

$$\ell_k = \log_{10}(\min\{\max(g_k, 10^{-12}), 10^{20}\}), \quad k \in \{s, q, b\}. \quad (11)$$

After the relevant fold-specific selector has been fitted, Stage B executes each selected Skip, Query, and Behavior-only path from the same decision state to termination three prespecified times under a cyclic path order. These replays determine timing only; they never replace the Stage-A scientific endpoints. Each repetition retains its status, effective FE, timeout, completion, failure fields, order, component times, and full future-path time. For each repetition, the censored time equals raw observed time when the path completed and equals the larger of raw observed time and the role-specific timeout when the path timed out or failed. Let T_s, T_q, T_b be the medians of the three prespecified censored future-path times; the raw observed medians are diagnostic only. When the Stage-A path completed, every completed replay must reproduce its gap, observed target hit, and first hit. When Stage A did not complete, completed replays are checked only against one another and never replace the failed scientific endpoint. Planned FE must retain the path identity. Replay effective FE is retained per repetition and need not equal Stage A or another repetition; it never replaces the Stage-A scientific effective FE. Path-identity consistency, completed-replay internal endpoint consistency, and Stage-A-to-completed-replay consistency are stored separately. Within-Stage-B status instability and Stage-A/Stage-B completion instability are also distinct. A legacy failure-worst-case field, if retained, is a strict alias of the censored value. A timing or status observed after execution cannot be used to selectively rerun, delete, or replace a repetition. The Query time begins at the decision state and includes sample generation and objective evaluation, descriptor computation, downstream selection, any handoff, and continuation; the other two times include every operation their paths perform after that same state. The already executed common prefix is a sunk cost and is excluded from all three T_k . Repeating isolated action components does not substitute for these selected state-to-terminal replays.

For policy-level resource reporting, W_π separately denotes the censored median Stage-B wall-clock over three prespecified executions from $FE = 0$ to termination under policy π , including its prefix, streaming Behavior and Decision inference, and any triggered downstream path. Its performance and FE endpoints remain those of the corresponding prespecified Stage-A scientific execution. A pre-run AAS policy also starts this measurement at $FE = 0$. W_π evaluates complete online execution, whereas T_k evaluates a choice conditional on reaching S_i ; neither quantity substitutes for the other.

3.4. Three distinct utility quantities

The joint net utility of acquiring the query and using its full selector is

$$U_q^{\text{joint}}(S_i) = (\ell_s - \ell_q) - \lambda_T(\log_{10} T_q - \log_{10} T_s), \quad (12)$$

where the primary setting is $\lambda_T = 1$ and $\lambda_M = 0$. Thus one decimal order of terminal-gap change and one decimal order of state-to-terminal runtime change receive equal scalarization weight; this is a study setting, not a universal resource preference.

The corresponding Behavior-only full-budget utility is

$$U_b(S_i) = (\ell_s - \ell_b) - \lambda_T(\log_{10} T_b - \log_{10} T_s), \quad (13)$$

and the operational Query increment over that path is

$$I_q(S_i) = (\ell_b - \ell_q) - \lambda_T(\log_{10} T_q - \log_{10} T_b) \quad (14)$$

$$= U_q^{\text{joint}}(S_i) - U_b(S_i). \quad (15)$$

I_q includes query evaluations, sample-best contribution, unequal continuation budgets, acquisition time, and the difference between the two fitted selectors. It is an operational path comparison, not a pure-information or causal estimand. We report it both over all eligible states and over states first triggered by the proposed policy. If $U_q^{\text{joint}} > 0$ but $I_q \leq 0$ in the corresponding population, the evidence supports the joint Query/full-selector path over SBS, but not query acquisition over the full-budget Behavior-only path.

The marginal predictive contribution of query descriptors is a separate diagnostic. A state-only selector and the full Query selector use the same query-budget four-action outcomes; their OOF-selected continuation-only gaps define

$$\Delta_{q,\text{pred}} = \ell_{\text{state-only selected,cont}} - \ell_{\text{full query selected,cont}}. \quad (16)$$

This comparison adds no action runs, excludes the sample best, and does not charge acquisition cost. It cannot replace Eq. (15). Because the protocol does not include a matched-acquisition state-only fourth path, neither comparison separately identifies the operational contribution of the descriptors from sampling, sample-best, FE, runtime, and budget effects.

The continuous joint utility is retained, with binary target

$$y_q(S_i) = \mathbf{1}[U_q^{\text{joint}}(S_i) > 0]. \quad (17)$$

The Decision input contains only information available before the query. It excludes query descriptors, benchmark and algorithm identity, optimizer parameters, known-optimum gaps, action outcomes, and future states.

3.5. First-trigger policy

For run r , opportunities are ordered by exact FE and opportunity index. The deployed trigger is

$$t_r^* = \min\{t : z_\theta(S_{r,t}) > \tau_{\text{OOF}}\}, \quad (18)$$

with no trigger when the set is empty. A run can trigger at most once, and states after t_r^* are unreachable under that policy. The threshold is estimated only from BBOB-training scores produced by a fold-specific upstream OOF chain; BBOB validation and external suites do not influence it. Ridge produces a continuous utility prediction, whereas LDA and logistic regression produce classification scores that are not interpreted as calibrated utility magnitudes.

3.6. Action-loss decomposition and claim boundary

Within the relevant action matrix, define

$$L_{\text{best}}(S_t) = \min_{a \in \mathcal{A}(S_t)} L(S_t, a). \quad (19)$$

Then

$$V_{\text{potential}}(S_t) = L_{\text{skip}}(S_t) - L_{\text{best}}(S_t), \quad (20)$$

$$R_{\text{selector}}(S_t) = L_{\text{selected}}(S_t) - L_{\text{best}}(S_t), \quad (21)$$

$$L_{\text{skip}}(S_t) - L_{\text{selected}}(S_t) = V_{\text{potential}}(S_t) - R_{\text{selector}}(S_t). \quad (22)$$

Population-transfer effects already appear in the observed action outcome and state-to-terminal future-path runtime; query FEs already reduce the continuation budget. They are not charged again.

The resulting estimands are specific to the evaluated query, portfolio, selector, transition rule, budget, and state distribution. Shared-state action runs do not identify a causal effect. The formulation does not presuppose that most states have non-beneficial utility, that Behavior predicts utility, or that the policy transfers to an external suite.

4. Decision-before-Feature method

4.1. Overview

Decision-before-Feature is trained offline. Native optimizer trajectories are first collected under a fixed sampling protocol. Each accepted state is converted into a permutation-invariant Behavior vector available before landscape descriptors are computed. Stage A runs each query-adjusted and full-budget four-action outcome once to supervise fold-specific Query and Behavior-only selectors and to fix the scientific gap, observed first hit, target hit, path completion, completion-conditioned endpoint success, planned FE, effective FE, and failure status. Stage B freezes their selected actions and executes the selected Skip, Query, and Behavior-only state-to-terminal future paths three prespecified times under cyclic path order. A completed replay retains its raw observed time, whereas a timed-out or failed replay contributes the larger of its raw time and role-specific timeout; the replays provide the median of these three censored conditional wall-clock values in Eqs. (12)–(15). The raw median is diagnostic only. The performance and FE terms remain those of Stage A, and the already executed shared prefix is excluded. Repeated action components do not replace selected future paths. Every Stage-B repetition retains its status, observed hit, effective FE, timeout, and completion. Path identity, consistency among completed replays, and Stage-A-to-completed-replay consistency are checked separately, as are within-Stage-B status instability and Stage-A/Stage-B completion instability. Repetitions are never selected or rerun after their time or status is observed. A lightweight

Decision Model then maps pre-query Behavior alone to a query score. The deployed controller performs no online parameter update, and its separate FE-zero-to-terminal wall-clock follows the same separation between a Stage-A scientific execution and three Stage-B timing replays.

Figure 1 summarizes the separation between offline data construction, query-specific downstream selection, decision-model freezing, and online execution. The use of trajectory-derived summaries follows the general observation that optimizer histories carry information about search dynamics and problem–algorithm interaction [? ? ?]; their value for the present query decision remains an empirical question.

This separation gives the two supervised components different roles. The *selection reference* operates after query acquisition and may use the query descriptors. The *decision model* operates before query acquisition and is prohibited from using query descriptors, function identity, dimension, algorithm identity, optimizer-specific parameters, action losses, or best-observed-action diagnostics.

4.2. Native-state trajectory and window alignment

For every portfolio optimizer, a complete state contains its population, objective values, best-so-far solution, generation or native-update count, internal dynamic variables, and random-number-generator state. Restoring this state continues the same optimizer rather than constructing a new run from population values. This distinction is essential for PSO velocities, CMA-ES distribution parameters, and the adaptation histories of DE and SHADE.

Let e_t be the actual FE count at an accepted state and let $w \in \{0.02, 0.05, 0.10\}$ be a nominal behavior window. Among the complete native updates preceding t , the anchor is

$$a(t, w) = \arg \max_u \{e_u : e_u \leq e_t - \text{round}(wB)\}. \quad (23)$$

For population size $N = 40$, the realized span satisfies

$$\text{round}(wB) \leq e_t - e_{a(t,w)} < \text{round}(wB) + N. \quad (24)$$

Rates and slopes use the realized FE-ratio span, not the nominal value. The realized ratio, FE count, and number of native updates are retained as measurement metadata and are excluded from the decision input.

Behavior vectors are evaluated only at accepted complete-update states. The budget milestones, behavior-triggered opportunities, rearming rules, and per-phase quotas are specified in Section 5; opportunity labels determine state sampling but are excluded from the Decision Model.

4.3. Permutation-invariant behavior representation

For distance-based population summaries, coordinates are mapped to the unit hypercube using the problem bounds,

$$\tilde{x}_{ij} = \frac{x_{ij} - \ell_{pj}}{u_{pj} - \ell_{pj}}. \quad (25)$$

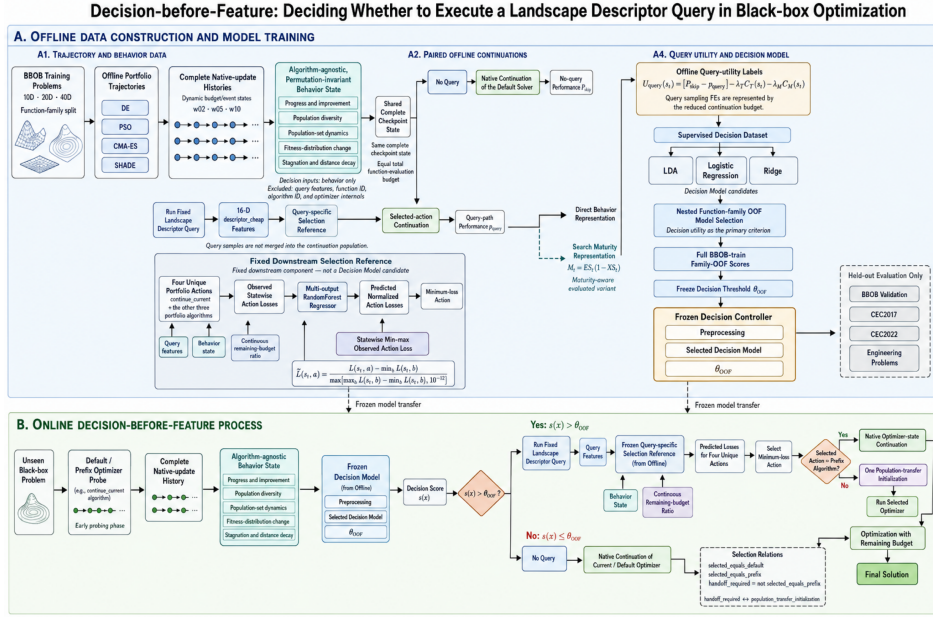


Figure 1: Decision-before-Feature protocol. The upper panel constructs state-aligned joint, Behavior-only, and operational-increment labels and freezes the downstream Selection References, Decision Model, preprocessing, and threshold using BBOB training function groups. The lower panel applies those frozen components at an online decision opportunity. Query descriptors enter only the downstream Selection Reference; the pre-query Decision Model receives algorithm-agnostic behavior.

This unit-cube convention defines distance normalization. No correspondence between row i at two checkpoints is assumed. The current population diversity is the average pairwise distance

$$D_t = \frac{2}{N(N-1)} \sum_{1 \leq i < j \leq N} \|\tilde{x}_i^{(t)} - \tilde{x}_j^{(t)}\|_2. \quad (26)$$

For an anchor population and the current population, the empirical assignment distance is

$$W_t^{(w)} = \frac{1}{N} \min_{\pi \in S_N} \sum_{i=1}^N \|\tilde{x}_i^{(a)} - \tilde{x}_{\pi(i)}^{(t)}\|_2, \quad (27)$$

and its rate divides Eq. (27) by $(e_t - e_a)/B$. Centroid shift and its rate are computed analogously. Centroid-shift coherence is the ratio of centroid displacement to $W_t^{(w)}$, clipped to $[0, 1]$ and set to zero when the population does not change. Covariance trace and effective-rank changes use the unit-cube populations. Current covariance spectral concentration is computed from the raw-coordinate covariance. For its nonzero eigenvalues λ_j and $r = \min(d, N-1)$,

$$C_t = \frac{r \sum_j \lambda_j^2 / (\sum_j \lambda_j)^2 - 1}{r - 1}, \quad (28)$$

with $C_t = 0$ for a degenerate covariance or $r \leq 1$.

Fitness changes use a shift-invariant scale. Let

$$I_0 = \max(\text{IQR}(y_{\text{init}}), 10^{-12}), \quad (29)$$

where y_{init} is the evaluated population immediately after optimizer initialization and before any native update. Improvement rate over a short window is

$$IR_t^{(w)} = \frac{f_{\text{best}}(a) - f_{\text{best}}(t)}{I_0 (e_t - e_a) / B}. \quad (30)$$

Improvement frequency is the proportion of native-update intervals in which best-so-far fitness decreases strictly. To compare fitness distributions without individual identities, the two endpoint vectors are sorted. If $y_{(i)}^{(a)}$ and $y_{(i)}^{(t)}$ denote their empirical quantiles, the method records the fraction for which $y_{(i)}^{(a)} - y_{(i)}^{(t)}$ exceeds the numerical strict-improvement tolerance, their mean signed improvement divided by I_0 and the realized ratio span, and their mean absolute difference on the same scale.

Stagnation is based on the last strict best-so-far improvement. With e_{last} denoting its FE count,

$$S_t^{(0.10)} = \frac{\min\{\max[(e_t - e_{\text{last}})/B, 0], 0.10\}}{0.10}. \quad (31)$$

Distance decay compares the population's mean unit-cube distance to its current population-best at the long-window endpoints. Convergence rate is the relative decrease in D_t divided by the realized FE-ratio span. The reference point in these calculations is the current population-best, not a known global optimum.

The representation borrows the general idea that longitudinal population summaries can characterize problem-algorithm interaction [?], but uses one fixed-dimensional definition across all four optimizers and predicts query utility rather than a problem class. Distance-decay and stagnation

Table 2

Frozen behavior groups used in the feature-group comparison.

Group	Count	Added information	Role
T0	1	Actual FE ratio only	Time-only controller
B1	19	Progress, diversity, set and fitness-distribution change, stagnation, convergence, and elite concentration	Core permutation-invariant behavior
B2	25	B1 plus fitness-spread slope, population and elite centroid shifts, covariance trace ratio, covariance effective rank, and diversity recovery	Longitudinal set dynamics
B2+Motion	28	B2 plus three set-motion summaries	Motion-specific sibling ablation
B2+Maturity	28	B2 plus three deterministic maturity basis functions	Maturity-specific sibling ablation
B3	31	B2 plus both Motion and Maturity variables	Primary full input

semantics are also informed by behavior-characterization studies [? ?]. Measures requiring a known optimum, persistent particle identities, or an optimizer-specific interaction graph are excluded; in particular, particle-wise directional entropy is not an input.

The behavior representation has 34 unique outputs: 31 model-eligible variables and three diagnostic-only variables. The nested groups are summarized in Table 2.

The raw fitness standard deviation, population-overlap statistic, and best-distance–fitness correlation are diagnostic only. In particular, the last quantity is close to a local landscape proxy and is excluded from the pre-query model.

4.4. Operational search-maturity variables

Search Maturity denotes three deterministic, study-specific basis functions of existing Behavior, rather than an observed state, established convergence measure, latent variable, or causal mediator. Define

$$h_+(z) = \frac{\max(z, 0)}{1 + \max(z, 0)}, \quad h_-(z) = \frac{1}{1 + \max(z, 0)}, \quad (32)$$

$$c(z) = \min\{\max(z, 0), 1\}.$$

Using the defined behavior variables, exploration stabilization is the mean of

$$h_+(-\dot{D}_t), \quad h_-(\dot{W}_t), \quad c(C_t), \quad c(K_t), \quad (33)$$

where $\dot{D}_t$ is the medium-window diversity slope, $\dot{W}_t$ is the population assignment-distance rate, and K_t is centroid-shift coherence. Exploitation saturation is the mean of

$$c(S_t), \quad h_-(IR_t), \quad h_+(DD_t), \quad h_+(CR_t), \quad h_-(E_t), \quad (34)$$

where DD_t , CR_t , and E_t denote distance decay, convergence rate, and elite-to-population diversity ratio. The primary maturity variable is

$$M_t = ES_t(1 - XS_t). \quad (35)$$

The B3 group also contains the bounded linear alternative $c[(ES_t + 1 - XS_t)/2]$ and an exploration-to-exploitation ratio. Undefined components remain missing and are handled by training-only median imputation. The three variables

add no raw information; their predictive contribution is assessed only through the prespecified B2+Maturity – B2 and B3 – (B2+Motion) contrasts. No monotone or non-monotone relationship with Utility is built into the definition.

4.5. Downstream action-loss selectors

For every eligible shared state, all actions in Eq. (6) are run once under both budgets in Eqs. (7) and (8). Their targets use the within-state transform in Eq. (9). The fixed estimator is a multi-output random-forest regressor with 200 trees, minimum leaf size two, training-fold median imputation, and random seed 1701.

The full Query selector predicts the four query-budget losses from B3 Behavior, the current query descriptors, and continuous query-adjusted remaining budget,

$$\hat{L}^q(S_t) = g_q\left(b_t, \phi_q, \frac{B - e_t - FE_q}{B}\right), \quad (36)$$

$$\hat{a}_t^q = \arg \min_a \hat{L}_a^q(S_t). \quad (37)$$

A query-adjusted state-only selector omits ϕ_q while using the same outcomes and budget; its OOF-selected continuation-only gap supplies Eq. (16). The Behavior-only selector uses B3 Behavior and $(B - e_t)/B$ to predict the distinct full-budget matrix. Traditional pre-run AAS uses a fourth, query-only selector at $e_t = 0$ and no trajectory Behavior. These fitted references are downstream experimental components, not the proposed contribution.

Every selector prediction used for training evidence is function-cross-fitted. Within an outer Decision fold, selectors are refitted only on outer-fit functions; within every inner fold, they are again refitted only on inner-fit functions. A complete-training selector is used only after OOF evidence has been produced, for BBOB validation and external deployment. A selector fitted on all BBOB-training functions cannot generate its own OOF Utility labels.

For each selected action, three relations distinguish whether it equals the default, whether it equals the prefix, and whether a handoff is required. Writing h_t for the handoff indicator, i_t for equality with the prefix, and r_t for

the transition type, these relations satisfy

$$h_t = \neg l_t^{\text{prefix}} = \mathbf{1}[r_t = \text{population transfer initialization}], \quad (38)$$

Here h_t , l_t^{prefix} , and r_t correspond, respectively, to the reported handoff indicator, equality with the prefix action, and transition type. Equality with the default action is reported independently. These relations hold by definition for every state, support stratified reporting, and are not Decision Model inputs. The fields `selected_equals_default`, `selected_equals_prefix`, `handoff_required`, and `handoff_type` remain distinct; no generic same/changed-algorithm alias replaces them.

4.6. Decision-model selection and first-trigger thresholds

The active candidates are fixed to (i) linear discriminant analysis with default estimator settings, (ii) logistic regression with $C = 1$, balanced class weights, and the L-BFGS solver, and (iii) Ridge regression with $\alpha = 1$. Each is a pipeline with training-fold median imputation and standard scaling. LDA and logistic regression fit Eq. (17); Ridge fits the continuous U_q^{joint} . The model-family comparison uses B3 only.

For both Selector and Decision fitting, the primary sample weights are computed inside each fit fold. Functions receive equal total weight, as do fixed dimension strata within a function, static problems within a function–dimension cell, and optimizer runs within a static problem. A run’s weight is divided equally among its eligible states, and the resulting row weights are normalized to mean one within the fit fold. Holdout rows and outcome direction cannot affect these weights. Unit state-row weighting is a prespecified sensitivity analysis. The current estimators have not yet been wired to this weighting protocol, so model freezing remains blocked until that implementation is complete.

The entire learning chain is nested within BBOB training. For each outer function holdout, only outer-fit functions determine SBS_{outer} , cross-fitted Query and Behavior-only selectors, Utility labels, preprocessing, Decision fits, and the outer threshold. Within every inner function fold, the procedure again recomputes SBS_{inner} , refits/cross-fits the selectors, regenerates Utility, and fits preprocessing and the Decision Model using inner-fit functions only. Concatenated inner-holdout scores determine the outer first-trigger threshold. Frozen outer components are then evaluated once on the outer holdout. A label table generated by full-training selectors followed by Decision-only OOF does not satisfy this design.

For a run r with ordered opportunities $j = 1, \dots, K_r$, define

$$J_r(\tau) = \min\{j : z_{rj}^{\text{OOF}} > \tau\}. \quad (39)$$

An empty set contributes zero Utility; otherwise the run contributes only $U_{q,rJ_r(\tau)}^{\text{joint}}$. The threshold

$$\tau_{\text{OOF}} \in \arg \max_{\tau} \text{Mean}_{\text{run} \rightarrow \text{problem} \rightarrow \text{function}} \left[\mathbf{1}\{J_r(\tau) \text{ exists}\} U_{q,rJ_r(\tau)}^{\text{joint}} \right] \quad (40)$$

uses equal weighting at each level. Ties prefer fewer called runs and then the larger numerical threshold.

Candidate quality is the function-balanced first-trigger mean Utility after concatenating outer holdouts. Ties use the prespecified order LDA, logistic regression, then Ridge. AUROC, average precision, Spearman correlation, Ridge RMSE, T0, and BBOB validation do not alter this choice.

After B3 freezes the model family, the same named estimator is used for T0, B1, B2, B2+Motion, B2+Maturity, B3, and Behavior-only fitting. Each group obtains its own train-only first-trigger threshold without reselecting the model or a preferred input group. The primary T0 comparison uses only the 12 mandatory milestones and recomputes B3 on the same rows; a time-only model on the Behavior-triggered schedule is reported only as schedule-conditioned sensitivity. A second static-context diagnostic, *dimension-stratified T0*, fits a separate FE-ratio-only controller within each evaluated dimension. It does not add dimension to the primary Decision input and cannot replace the milestone-matched B3–T0 comparison.

A final full-training OOF pass repeats the same fold-specific upstream chain to freeze the deployable Query and Behavior-only thresholds. It also freezes the matched-rate Random run-call probability and empirical first-trigger FE-ratio distribution. Only afterward are final components refitted on all BBOB training functions. BBOB validation and external suites are evaluation-only.

At deployment, the default SBS advances natively while the streaming Behavior extractor observes complete updates. If the score does not cross the frozen threshold, native optimization continues. At the first crossing, the fixed query is executed once, the full Query selector chooses an action, and its selected future path consumes the remaining query-adjusted budget. There is no subsequent trigger, online training, or query-type selection.

5. Experimental setup

5.1. Research questions and evaluation design

The empirical study addresses the five prespecified questions in Table 3. Each question is linked to a target distribution, a comparison set, and uncertainty summaries that respect the nested dependence of decision states within runs, problems, and benchmark functions. A benchmark function, rather than a state, run, transformed instance, or function–dimension combination, is the top-level statistical unit.

5.2. Internal benchmark and grouped-by-function partition

The internal benchmark is the noiseless BBOB suite. The training set comprises functions 1, 2, 3, 4, 6, 7, 8,

Table 3

Research questions and corresponding evaluation criteria.

RQ	Question	Evaluation criteria
RQ1	Statewise value of the primary query	State–run–static–problem–fixed–dimension–stratum–function–balanced Utility distribution and decomposition; proportion with $U_q \leq 0$; effect intervals.
RQ2	Information beyond elapsed budget	Grouped-by-function training OOF candidate selection and diagnostics; the prespecified milestone-matched B3–T0 contrast is reported as a previously inspected BBOB-validation internal estimate and separately on each eligible external suite.
RQ3	End-to-end policy trade-off	First-trigger call quality, final performance, FE use, runtime, and paired comparisons with all required policies.
RQ4	Transport of the frozen procedure	Separate previously inspected BBOB and CEC2017 development estimates and prospective CEC2022 and engineering estimates, including coverage and prespecified failures.
RQ5	Behavioral associations with query value	Milestone-matched behavior increments, maturity-separated ablations, parameter stability, and prespecified strata.

10, 11, 12, 15, 16, 17, 18, 20, 21, 22, and 23; functions 5, 9, 13, 14, 19, and 24 form the function-held-out validation split. Functions, rather than randomly selected transformed instances, define the grouping variable. This partition addresses evidence that instance-level splits can overstate algorithm-selection generalization [?]. Both sets use dimensions $d \in \{10, 20, 40\}$, instances 1, 2, 3, optimizer seeds 1, ..., 30, population size 40, and total budget $B = 1000d$. The portfolio contains DE, PSO, CMA-ES, and SHADE, giving 54 training and 18 validation function-dimension combinations.

The single best solver (SBS) is estimated exclusively from terminal outcomes at $FE = B$. Each raw gap is clipped with the suite endpoints and transformed to $\log_{10}$ gap. For each algorithm, outcomes are averaged with equal weight from run to problem (dimension–instance pair) to function; the algorithm with the smallest function-balanced mean is selected. A tie is resolved in the prespecified order DE, PSO, CMA-ES, SHADE. The procedure is repeated using only the fit functions of every outer and inner fold. It does not use average ranks, and intermediate decision states do not contribute.

BBOB validation is a previously inspected internal development-evaluation set. Historical model comparison, tuning, and ablation work examined its outcomes, so excluding it from the current query selection, preprocessing, model-family selection, input-group comparison, threshold fitting, and score-neighborhood definition does not restore independent-confirmation status. Its six functions are six top-level evaluation units. Its role is estimation-first: all six function-level effects, the fixed-set mean, and conditional intervals are reported, but its results do not trigger model reselection or binary claims based on a p -value threshold.

5.3. External benchmarks

External evaluation is organized as non-pooled, suite-specific analyses. The current CEC2017 configuration specifies dimensions 10, 30, and 50, seeds 1, ..., 30, population size 40, and budget $B = 1000d$. Its function list is not yet an executable manuscript commitment: the current F1–F29 list includes F2 and excludes F30, while the project contains no source establishing that choice for the implementation in use. The F2/F30 treatment must be reconciled with the implementation and benchmark definition before any further evaluation under the current protocol. The four-algorithm portfolio and dynamic opportunity protocol remain unchanged, and all learned components are then held fixed at their BBOB-training estimates.

CEC2022 and engineering problems form separate components of RQ4. Their problem sets, dimensions, bounds, budgets, target errors, error floors and caps, wall-time limits, and official constraint-handling rules are specified before any policy outcomes are inspected. A run that raises an optimizer exception, returns a non-finite objective value, exceeds its fixed wall-time limit, or violates a suite-defined completion condition is retained as a run failure. A target hit observed before that failure remains a target hit for standard ERT, while path completion is false and completion-conditioned endpoint success is false. A failed path with no observed hit contributes its full planned budget to ERT. The path receives the prespecified suite-level worst rank and capped error for summaries that require a finite value; the cap is never estimated from observed policy contrasts.

For every external suite, an attempted query whose descriptor group cannot be computed remains in the coverage denominator and is recorded as a query failure. The evaluations and time already spent on that query remain charged, no replacement descriptor group is invented, and the optimizer continues the prefix action with its remaining budget. Individual undefined descriptor values may be imputed

Table 4

Prespecified landscape-descriptor queries.

Query	Sample	FE share	Features	Role
descriptor_cheap_invariant	LHS 50 <i>d</i>	5%	14	Primary
pflacco_standard_invariant	Same LHS 50 <i>d</i>	5%	37	Configuration robustness
pflacco_broad_invariant	Independent LHS 100 <i>d</i>	10%	52	Configuration robustness

only with frozen BBOB-training medians; a group-level extraction failure is not converted into a successful query by imputation. Missing benchmark implementations or unexecuted problems are reported as missing coverage and cannot be counted as completed evaluation. Previously inspected BBOB validation, CEC2017, prospective CEC2022, and engineering estimates are reported separately; neither partial coverage nor agreement on a subset of suites is interpreted as unconditional cross-benchmark transfer.

For BBOB training/validation and CEC2017, the prespecified values are a raw-gap floor of 10^{-12} , a raw-gap and failure-loss cap of 10^{20} , success target 10^{-8} , and a 3600-second state-action timeout. First hits are checked after every objective evaluation, including evaluations in a query sample. A timeout is retained as path failure. It does not erase an earlier observed hit; standard ERT uses that hit, whereas an unhit path contributes its full planned budget. Path completion and completion-conditioned endpoint success are reported separately. CEC2022 and engineering problems cannot be run until analogous endpoints and their constraint-handling rules have been fixed.

For every suite and endpoint, the attempted denominator and coverage are shown beside the complete-pair estimate. Unclosed pairs receive two prespecified extreme assignments: gap uses the suite floor and cap; target hit uses one and zero; an unsuccessful ERT contribution uses the full budget; adverse runtime uses the timeout; and favorable runtime uses the smallest strictly positive runtime among that suite's complete pairs, with this rule frozen before a run. Utility is recomputed from the same extreme gap and runtime endpoints rather than assigned an arbitrary value. A scientific path failure already retained through a cap or timeout is an observed outcome, not a missing pair. If either extreme analysis changes the effect direction or the interval's position relative to the operational tolerance, or if coverage is below 95%, the corresponding conclusion is not established.

5.4. Landscape-query configurations

Table 4 defines the three prespecified queries. Primary inference concerns `descriptor_cheap_invariant` at $\lambda_T = 1$ and $\lambda_M = 0$. The two `pflacco` configurations are prespecified robustness analyses of representation and acquisition cost. They cannot replace the primary query after outcomes are observed, and agreement among them does not support a claim beyond the three evaluated configurations.

The cheap and standard queries use the same evaluated (X, y) sample. The cheap list excludes the response median and IQR because the common median/IQR preprocessing makes them constants zero and one. Their removal changes

neither the sample nor the action outcomes. Sharing the sample yields a paired comparison of descriptor representations at a common FE cost. The broad query uses an independent 100*d* design. The standard configuration comprises PCA, nearest-better clustering, dispersion, information-content, and ELA-distribution groups. The broad configuration adds ELA level-set and sample-derived fitness-distance-correlation groups, while excluding the full quadratic meta-model group, cell mapping, and feature groups that require additional objective evaluations. Query sample points are not inserted into the optimizer population. Each query defines a separate downstream selector, utility target, decision model, threshold, baseline comparison, and external evaluation; observations from different query identities or FE designs are not pooled.

Each static problem and sample design has one problem-keyed LHS realization, shared across optimizer seeds, decision states, and actions. The primary estimands are therefore conditional on these fixed query realizations; without multiple independent LHS replicates they do not average over query-sampling randomness. Additional replicates could only enter as a prespecified sampling-robustness sensitivity, with replicate identifiers included in pairing and hierarchical analysis. Standard versus broad changes representation, sample size, sample realization, and action budget simultaneously, so that contrast is interpreted only at the whole-query-configuration level.

5.5. Dynamic decision opportunities

Offline label construction and policy evaluation use the same dynamic budget-event sampling scheme. Complete native updates are monitored when they cross a grid from 0.20*B* to 0.60*B* in increments of 0.01*B*. The mandatory budget milestones are 0.20, 0.22, 0.24, 0.26, 0.28, 0.30, 0.34, 0.38, 0.42, 0.46, 0.50, and 0.60. Five event types may add decision states: resumed strict improvement; a stagnation span reaching 0.05*B*; an absolute medium-window covariance-effective-rank change reaching 0.20; unit-cube elite-centroid displacement divided by $\sqrt{d}$ reaching 0.05; and relative diversity recovery reaching 0.10. Covariance-effective-rank-change, elite-migration, and diversity-recovery events rearm when their respective quantities fall below 0.10, 0.025, and 0.05. Stagnation rearms after a new strict best-so-far improvement, whereas resumed improvement rearms when short-window improvement frequency returns to zero.

Each complete update that crosses one or more monitor points is evaluated once. If a crossing includes a milestone, the event is associated with that milestone and does not

consume an event-only quota or spacing anchor. Otherwise, the most recent crossed monitor point supplies the nominal location. Each of the early, middle, and late phases admits at most two event-only states, with at least $0.02B$ separation in realized FE ratio. Suppressed crossings still update event rearming. This scheme admits 12–18 states per run. All accepted states are retained for the full B3 controller, but they are not treated as independent or globally equal-weight observations. The realized FE/B represents elapsed budget and is the sole T0 covariate; exact integer FE aligns shared-state outcomes. Nominal milestones and native-update counts describe the sampling process but do not add Decision Model inputs. T0 is fitted, thresholded, and evaluated only on the twelve mandatory milestone states. Whenever B3 is compared directly with T0 as an information ablation, the B3 estimate is recomputed on the same milestone rows; event-only opportunities therefore cannot disclose behavioral event occurrence to the time-only comparator.

5.6. Matched states and analysis population

The analysis pairs all alternatives at an identical complete native state. For each eligible state, it combines permutation-invariant behavior, the shared $50d$ and independent $100d$ LHS samples, the three descriptor representations, and the query-budget-specific outcome of every eligible action. A trajectory enters an estimand only when both its complete native history and terminal outcome satisfy the same state definition. Query-specific selection references, utilities, decision models, and policies are estimated from these matched observations under the function-group partitions described above.

The primary Decision Model population contains states for which both the prefix and default algorithm equal the training SBS and the no-query path continues that optimizer. States generated from the other prefix algorithms are reserved for cross-prefix robustness, leave-one-prefix-algorithm-out analysis, and tests of portability across the four search processes. They are excluded from primary model fitting, threshold fitting, and primary result aggregation.

Offline generation is explicitly two-stage. The four-action matrices and $FE=0$ four-algorithm outcomes are each generated once to fit fold-specific selectors. These prespecified Stage-A executions also fix terminal gap, observed first hit, target-hit status, path completion, completion-conditioned endpoint success, planned FE, effective FE, and scientific failure status. Those selectors emit a replay plan; only then are their selected Skip, Query, and Behavior-only paths from the corresponding decision state to termination executed three prespecified times with cyclic path order. The pre-run AAS and complete online-policy paths use the same division: one Stage-A scientific execution and three Stage-B timing replays from FE zero. Stage B provides only wall-clock measurements and repetition-level status, effective FE, timeout, completion, and failure diagnostics; it cannot overwrite a Stage-A

scientific endpoint. A completed replay contributes its raw observed time to the censored time, whereas a timed-out or failed replay contributes the larger of its raw time and role-specific timeout. Utility uses the median of the three censored times; the raw observed median is diagnostic only. Path-identity consistency, consistency among completed replays, and consistency between Stage A and completed replays are stored separately, as are within-Stage-B status instability and Stage-A/Stage-B completion instability. No repetition is selectively rerun after its time or status is observed. Utility is constructed after these replays from Stage-A performance and Stage-B state-conditional future-path times for Skip, Query, and Behavior-only. The replay plan is a scientific dependency, not an optional timing pass: without an offline decision-state-to-terminal producer and a fold-role-complete plan, formal data generation does not begin.

5.7. Comparison policies

Nine prespecified comparison roles are retained because they answer different benchmarking questions. Every sequential decision policy is a first-trigger policy: it scans its eligible opportunities in increasing exact FE order, can trigger at most once in a run, and contributes only the Utility and terminal outcome associated with that first trigger. Once a query or behavior-only handoff has been initiated, later states in that run are not additional calls, observations, or policy rewards.

- **Never Query:** continue the complete SBS state natively and incur no query acquisition cost.
- **Always Query:** execute the fixed query at the first accepted decision opportunity and invoke the same downstream selector used by the proposed policy.
- **Matched-rate Random Analysis:** freeze the proposed controller's run-level call rate and empirical first-trigger FE-ratio distribution from the complete BBOB-training fold-specific OOF chain. At the start of a run, one Bernoulli draw determines whether a call is planned; if so, a target ratio is drawn and the first eligible opportunity not earlier than that target is used. The run does not call when no such opportunity exists. Thirty explicit SeedSequence streams are averaged within the same scientific run before problem/function aggregation; they are not independent scientific replications. Neither a constant opportunity hazard nor a state drawn uniformly after observing the complete opportunity set is used.
- **FE-zero query-only, sample-isolated pre-run AAS:** before any portfolio optimizer update, acquire the fixed LHS query, compute its descriptors, and use a separately trained query-only selector to choose one portfolio algorithm for the remaining budget. The query consumes its configured FE and runtime, its sample neither initializes nor augments the optimizer population, and the selected algorithm starts

from its fresh native initialization with budget $B - FE_q$. The selector uses query descriptors and static remaining-budget information but no trajectory behavior. For explicit relation accounting, the prefix and selected algorithms are set equal, selected-equals-prefix is true, handoff-required is false, handoff type is fresh-optimizer-initialization, and the default/no-query algorithm is the corresponding fold SBS. This baseline is not identified with the statewide Always Query policy, and its interpretation is conditional on the repository's fixed implementations, parameters, and population size 40.

- **SBS:** use the training-derived static default defined above. In the primary population, the prefix and default are this same SBS, so its native complete-budget path coincides with Never Query; the separate name preserves its role as the standard algorithm-selection reference rather than introducing another online trajectory.
- **VBS:** for each static problem defined by function, instance, and dimension, average each algorithm's complete-budget clipped $\log_{10}$ gap over optimizer seeds, select the algorithm with the smallest mean (ties follow DE, PSO, CMA-ES, SHADE), and then retain that algorithm's paired seed-level outcomes for aggregation. It never chooses an algorithm separately for each seed. It is a non-deployable problem-level hindsight reference, does not necessarily upper-bound a dynamic policy run by run, and is distinct from the statewide best observed action.
- **Milestone-only Time Controller:** use the selected Decision Model family and the same training-only threshold procedure as the full controller, with input restricted to $\{FE/B\}$. It is fitted and checked only at the twelve mandatory milestones and stops after its first trigger.
- **Self-thresholded Behavior-only Selection:** fit the same frozen model family to U_b and obtain its own train-only first-trigger threshold. At its first crossing, the full-budget Behavior-only selector chooses the continuation or handoff action; query-call rate and query FEs are zero, while trigger and handoff rates remain reportable.
- **Decision-before-Feature:** use B3 Behavior with the selected model and the threshold estimated from full-training fold-specific upstream OOF predictions; acquire the query only at the first threshold crossing and invoke the full state-plus-query selector once.

In the primary population, Never Query and SBS yield the same native terminal outcome and are therefore reported once as *Never Query / SBS*. The other seven roles produce distinct outcome rows, giving eight distinct rows for the nine roles. Traditional pre-run AAS remains separate from Always Query because it queries before an optimizer trajectory

exists and commits to one algorithm. For every applicable endpoint, the shared Never Query / SBS outcome enters summaries and multiplicity counts once.

All methods receive the same total FE budget, portfolio, benchmark bounds, and suite-specific stopping rules. Sequential full-state policies use the same accepted opportunity stream, except that the time-only controller is deliberately restricted to milestones. Query policies use the same query definition and full downstream selector, and statewide handoffs use the same population-transfer rule. The pre-run AAS and behavior-only selectors are separate prespecified estimators because their available information and action budgets differ. A distinct matched-trigger Behavior-only diagnostic acts at the proposed controller's trigger solely to compute I_q ; it is not the self-thresholded baseline. Query-adjusted state-only, query-only, Behavior-only, and full selector regressions remain component diagnostics in addition to the policy-level baseline.

5.8. Evaluation measures

Decision-model family selection uses nested grouped-by-function OOF mean decision utility. AUROC, average precision, and Spearman correlation are secondary score-level measures. Continuous-utility RMSE is reported only for Ridge; classification probabilities and discriminant scores are not compared with continuous utility using RMSE.

Training outer OOF outcomes select among the three prespecified candidates and support development diagnostics only. Once the largest-OOF candidate is selected, the same OOF outcomes are not an unbiased estimate of the selected procedure. The prespecified RQ2 contrast—milestone-only B3 minus milestone-only T0—is reported on BBOB validation only as a post-development internal finite-set estimate, with each external suite reported separately. Candidate pairwise contrasts do not enter an RQ2 Holm family.

The primary fitting estimand gives equal weight successively to functions, fixed dimension strata, static problems, and optimizer runs, then divides each run's weight equally among its eligible states. Fold-local row weights are normalized to mean one and cannot depend on holdout rows or outcome direction. Unit weight per state row is retained only as a sensitivity analysis. The Decision and Selector estimators have not yet been wired to these cluster-balanced weights, which is a formal pre-run blocker; nested first-trigger evaluation does not by itself correct fitting weights.

Policy-level reporting includes run-call rate, first-trigger precision, mean joint decision Utility, matched-trigger I_q , and terminal outcomes. If J_r^π is policy π 's first trigger, its run Utility is

$$U_r^\pi = \begin{cases} U_q^{\text{joint}}(S_r, J_r^\pi), & J_r^\pi \text{ exists,} \\ 0, & \text{otherwise.} \end{cases} \quad (41)$$

The reported mean aggregates U_r^π equally from run to problem to function. Never Query has zero query-decision Utility by definition; the SBS role and VBS remain not applicable

for this quantity. Mean observed Utility conditional on calling is reported separately.

All sequential policies share one hindsight capture reference from the native SBS/default trajectory,

$$H_r = \max_t \max \{U_q^{\text{joint}}(S_{r,t}), 0\}. \quad (42)$$

Policy capture uses only its first trigger,

$$\text{utility capture}(\pi) = \frac{\sum_r \max \{U_q^{\text{joint}}(S_{r,J_r^\pi}), 0\}}{\sum_r H_r}, \quad (43)$$

with zero numerator for an untriggered run. Weighted numerator and denominator, their ratio, and the number of $H_r = 0$ runs are reported. Calls with $U_q^{\text{joint}} \leq 0$ are summarized by count, within-call proportion, and accumulated Utility loss. A state-score ranking diagnostic, if shown, is not called policy capture.

Policy-level endpoints comprise terminal clipped $\log_{10}$ gap, target-based success rate and expected running time (ERT), total and query FEs, FE-zero-to-terminal wall-clock time, behavior-extraction time, controller-inference time, query sampling and descriptor-computation time, downstream-selection latency, and peak memory. These full-run wall-clock values include the realized prefix and are reported separately from the decision-state-to-terminal future-path times that enter U_q^{joint} , U_b , and I_q . Success targets are suite specific and are fixed independently of the observed policy contrasts. Cost–performance reporting retains all nine prespecified comparison roles but uses eight distinct primary-population outcomes because Never Query and SBS share one native outcome. Traditional pre-run AAS remains distinct from Always Query.

5.9. Statistical analysis

Repeated states from one trajectory are not independent replicates. The aggregation hierarchy is state, run, static problem, fixed dimension stratum, and function. RQ1 first averages the complete eligible-state sequence within each run; policy endpoints start from one first-trigger outcome per run. Paired optimizer seeds are averaged within each static function–dimension–instance problem, static problems within each function–dimension stratum, and the fixed 10D, 20D, and 40D strata within each function. The benchmark function is the highest aggregation stratum.

The BBOB-validation estimand is the equally weighted finite-set mean for the six prespecified functions F5, F9, F13, F14, F19, and F24, conditional on those functions and the fixed dimensions and instances. It is a previously inspected post-development internal estimate; it neither treats the six functions or three transformed instances as probability samples nor supports inference to a function or transformed-instance superpopulation. The current choice of three instances and 30 optimizer seeds has no repository-local precision or power justification; it is a fixed development design, not evidence that a target half-width has been attained. Repetition counts for prospective CEC2022 and engineering

evaluations must receive an explicit precision target and be fixed before their outcomes are observed.

Primary intervals use 10,000 conditional paired hierarchical-bootstrap resamples. Every resample retains all six validation functions, every fixed dimension, and every static problem for instances 1–3; paired optimizer seeds are resampled only within each fixed static problem. An RQ1 seed/run retains its complete ordered state sequence as one cluster. Function resampling is reported only as a separately labelled function-composition sensitivity and never replaces the primary 95% interval.

ERT uses a dedicated paired ratio bootstrap rather than treating a run-level ERT contribution as an ordinary mean. Within each bootstrap replicate and each function–dimension cell, static problems remain fixed and paired optimizer runs are resampled within each problem. The FE numerator and target-hit count are recomputed for each policy, and the treatment-to-reference $\log_{10}$ ERT ratio is formed before fixed dimensions and functions are equally averaged. A one-sided zero-hit cell is retained as signed infinity and a two-sided zero-hit cell is undefined; neither is deleted, and either makes the corresponding interval unestablished. The suite-level consumer must combine this ratio analysis with attempted coverage and the two failure-sensitivity assignments before any suite conclusion is reported.

The numerical margins—mean Utility difference $[-0.01, 0.01]$, mean $\log_{10}$ -gap difference $[-0.05, 0.05]$, geometric-mean runtime ratio $[0.95, 1.05]$, and absolute call/target-hit-rate difference $[-0.05, 0.05]$ —are study-specific operational tolerances, not domain-grounded practical-importance bounds. Ordinary 90% intervals and Bonferroni simultaneous hierarchical-bootstrap intervals are descriptive sensitivity summaries of where each interval lies relative to these tolerances. For m contrasts, the latter uses two-sided level $1 - 2(0.05)/m$. Neither interval produces a confirmatory equivalence claim. Offset between gap and runtime inside Utility does not establish either endpoint, and lack of significance does not establish equivalence.

Prespecified paired contrasts use function-level sign-flip summaries under the null assumption that the six fixed function-effect signs are exchangeable. BBOB validation enumerates all $2^6 = 64$ sign vectors, so the smallest two-sided exact raw p value is $2/64 = 0.03125$. This summarizes sign consistency within the fixed set and does not create function-superpopulation inference.

Auxiliary Holm families are fixed separately by RQ and endpoint. RQ2 has one prespecified internal contrast, BBOB-validation milestone-only B3 versus T0; the three pairwise B3 candidate comparisons are selection diagnostics, not RQ2 family members. RQ3 compares the proposed policy with Never/SBS, Always Query, matched-rate Random, pre-run AAS, milestone-only T0, and self-thresholded Behavior-only; RQ5 contains the six prespecified feature increments. Suites and endpoints are not pooled. VBS, query-configuration and λ_T robustness, matched-trigger I_q , and

Table 5

RQ1 state-distribution summary. Utility is larger-is-better; $\log_{10}$ gap and runtime are smaller-is-better. Estimates aggregate state sequences within run, then run, problem, and function with equal weight.

Stratum	Coverage	$\Pr(U_q^{joint} \leq 0)$	Mean U_q^{joint}	Mean U_b	Mean I_q	Query-Skip $\Delta\ell$
BBOB-train outer OOF, all eligible states	—	—	—	—	—	—
Mandatory milestones	—	—	—	—	—	—
Event-only opportunities	—	—	—	—	—	—
BBOB validation, previously inspected	—	—	—	—	—	—

TBD: add 95% conditional hierarchical-bootstrap intervals, terminal and continuation-only gaps, sample-best contribution, first-hit offset, t

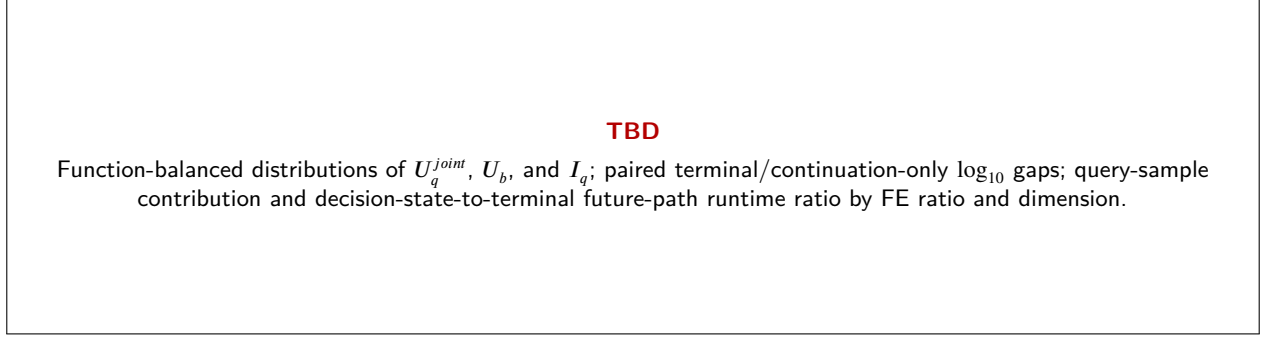


Figure 2: RQ1 distributional summary. Curves and bins are descriptive and do not treat repeated states as independent observations.

selector diagnostics are not members of these prespecified auxiliary Holm families. Analytically identical Never Query/SBS outcomes enter each summary and multiplicity count once.

On BBOB validation, the RQ3, RQ4, and RQ5 Holm families contain six, four, and six contrasts. Their smallest possible adjusted p values are therefore $6(2/64) = 0.1875$, $4(2/64) = 0.125$, and $6(2/64) = 0.1875$, respectively. None can reject at 0.05, regardless of the observed effect pattern. We report raw and adjusted p values only for transparency; function-specific effects, the fixed six-function mean, and its conditional interval are the primary evidence, and non-rejection is not interpreted as equivalence or absence of an effect.

6. Results

No formal result is available at the time of this protocol draft. Every dash in the following tables is an explicit missing value, not a zero. Values from the withdrawn population-reconstruction pipeline are not admissible here.

6.1. RQ1: What is the state-conditional value of the primary query?

RQ1 concerns `descriptor_cheap_invariant` with $\lambda_T = 1$ and $\lambda_M = 0$ on SBS-prefix states emitted by the prespecified dynamic opportunity process. It reports the joint Query utility, Behavior-only utility, and their operational increment together with terminal and continuation-only endpoints. Query terminal gap, observed target hit, and ERT include the best

point in the query sample; path completion and completion-conditioned endpoint success are reported separately. The sample contribution is also shown separately.

TBD The 10,000-resample conditional bootstrap estimates, which retain all six fixed validation functions, dimensions, and static problems and resample only paired optimizer seeds within each fixed problem, function-level effects, and prespecified sign-flip comparisons remain to be computed. The prevalence of $U_q^{joint} \leq 0$ is estimated without a predeclared majority direction.

TBD Any conclusion must state the fixed query realization, target state distribution, effect direction, endpoint-specific interval, and coverage. A joint advantage alone does not establish a descriptor-information advantage; if $I_q \leq 0$, it supports only the joint path over SBS. Both all-eligible-state and proposed-triggered I_q are required. Even when they agree, the absent matched-acquisition fourth path prevents descriptor-only operational attribution.

6.2. RQ2: Does Behavior add decision value beyond elapsed budget?

Model-family selection uses B3 only and the concatenated outer-holdout first-trigger U_q^{joint} . These outcomes select among candidates and provide development diagnostics; after selection they are not an unbiased estimate of the selected procedure. AUROC, average precision, and Spearman correlation are secondary; continuous-Utility RMSE is defined only for Ridge. After the model family is frozen, B3 is compared with T0 on the same 12 mandatory milestone rows. This prespecified BBOB-validation B3–T0 comparison is a post-development internal finite-set estimate, not an

Table 6

RQ2 nested model selection and milestone-matched Behavior–time comparison. Each outer/inner score is generated by fold-specific SBS, selectors, Utility, preprocessing, Decision fitting, and thresholding.

Panel	Evaluation	Model/input	First-trigger Utility	AUROC/AP	Spearman/RM
A	BBOB-train outer OOF	LDA / B3	–	–	– / N/A
A	BBOB-train outer OOF	Logistic Regression / B3	–	–	– / N/A
A	BBOB-train outer OOF	Ridge / B3	–	–	– / –
B	BBOB-train outer OOF milestones	Selected family / B3	–	–	–
B	BBOB-train outer OOF milestones	Selected family / T0	–	–	–
B	BBOB validation milestones	Selected family / B3	–	–	–
B	BBOB validation milestones	Selected family / T0	–	–	–
C	BBOB validation milestones	Selected family / dimension-stratified T0	–	–	–

TBD: add function effects, 95% CIs, call rates, selected model, and clearly labelled candidate-selection diagnostics.

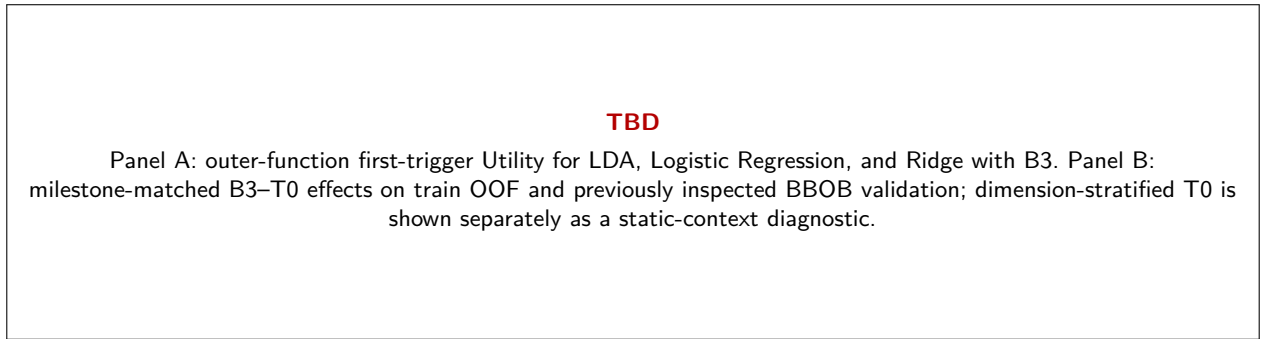


Figure 3: RQ2 model-family and Behavior–time comparisons. Validation is not used to select a model, input group, or threshold.

independent confirmation; external suites are evaluated separately. A dimension-stratified T0 fit, whose within-stratum input remains FE ratio only, is a static-context diagnostic and does not add dimension to the primary Decision input.

TBD The function-level sign-flip tests, hierarchical intervals, and candidate-selection record remain to be computed. BBOB validation has only six fixed functions and is interpreted through its six effects, finite-set mean, and conditional interval, not a significance-based reranking or function-superpopulation claim.

TBD Behavior provides value beyond elapsed budget only if the milestone-matched B3–T0 effect is supported on the primary policy endpoint and is not contradicted by terminal performance, runtime, or frozen evaluation. A schedule-conditioned T0 result cannot support that claim.

6.3. RQ3: What is the end-to-end policy trade-off?

All sequential policies make at most one first trigger. Never Query and SBS share one native outcome. Always Query and Traditional pre-run AAS are distinct: the former queries the first online state, whereas the latter queries at FE zero and starts one selected algorithm from native initialization. The self-thresholded Behavior-only policy has its own U_b threshold; the matched-trigger Behavior-only diagnostic used for I_q is not a baseline.

Matched-rate Random uses the training-OOF run-call rate and empirical trigger-FE distribution. Its 30 streams are averaged within each scientific run before statistical aggregation. Policy capture uses the same native-trajectory

run-level hindsight maximum for every sequential policy and only the first trigger in its numerator.

TBD Six proposed-versus-baseline contrasts are tested within each endpoint-specific Holm family: Never/SBS, Always, matched-rate Random, pre-run AAS, milestone-only T0, and self-thresholded Behavior-only. VBS and matched-trigger diagnostics are descriptive references. With six fixed BBOB validation functions, the two-sided exact sign-flip raw p value is at least 0.03125 and the Holm-adjusted value is at least 0.1875; rejection at 0.05 is impossible, so adjusted values are reported only for transparency.

TBD A resource-performance advantage requires explicit directions and intervals for terminal gap, runtime, calls, target hit, completion-conditioned endpoint success, and failure. Fewer query calls alone, or a favorable scalarized Utility produced by offsetting endpoints, is insufficient.

Table 7

RQ3 end-to-end outcomes. Eight non-duplicated outcomes represent nine comparison roles because Never Query and SBS coincide in the primary population.

Outcome (role)	Policy Utility	Terminal $\log_{10}$ gap	Future/full-run time ratios	Success	Query-call rate	Trigger rate
Never Query / SBS	0 / N/A	–	1.00	–	0	0
Always Query	–	–	–	–	–	–
Matched-rate Random Analysis	–	–	–	–	–	–
Traditional pre-run AAS	N/A	–	–	–	1	N/A
VBS	N/A	–	N/A	–	N/A	N/A
Milestone-only Time Controller	–	–	–	–	–	–
Self-thresholded Behavior-only	U_b : –	–	–	–	0	–
Decision-before-Feature	–	–	–	–	–	–

TBD: separate decision-state future-path and FE-zero-to-terminal policy times; add query/total FEs, ERT, first-trigger FE, first-call precision

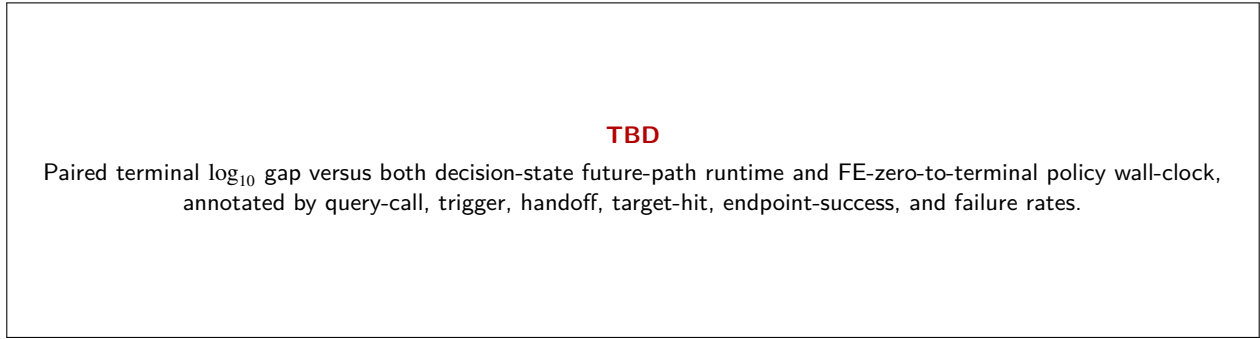


Figure 4: RQ3 cost–performance comparison. Utility is displayed alongside, not instead of, its performance and runtime endpoints.

6.4. RQ4: How does the frozen procedure transport beyond BBOB training?

BBOB validation, CEC2017, CEC2022, and engineering problems are separate evaluation domains. BBOB validation and CEC2017 are previously inspected internal and external development evaluations, respectively. CEC2017 remains blocked until the F2/F30 convention is reconciled. CEC2022 and engineering evaluations remain blocked until their functions/problems, dimensions, budgets, targets, caps, timeouts, and constraint-handling rules are fixed.

TBD Each executable suite requires its own conditional intervals, endpoint-specific Holm family, and failure sensitivity. BBOB validation reports its fixed six-function mean, all six function effects, and a formal CI that does not resample functions. Its four-contrast Holm family has minimum adjusted $p = 0.125$, so rejection at 0.05 is impossible.

TBD Transfer can be claimed only for a named, fully covered suite whose effect and failure sensitivity support it. Agreement on a subset of suites does not establish unconditional cross-benchmark generalization.

Table 8

RQ4 suite-specific evaluation of the fixed procedure. Undefined or unexecuted coverage is not converted into a completed result.

Suite	Definition status	Coverage	Mean Utility	Terminal $\log_{10}$ gap	Runtime ratio
BBOB validation	Previously inspected, 6 functions	–	–	–	–
CEC2017	Blocked: F2/F30 unresolved	–	–	–	–
CEC2022	Blocked: suite endpoints unresolved	–	–	–	–
Engineering problems	Blocked: problems/constraints unresolved	–	–	–	–

TBD: add attempted coverage, per-function/problem effects, query/selector/action/optimizer failures, ERT, two-direction extreme failure s

TBD Suite-separated function/problem effects for the frozen proposed policy and prespecified references, with coverage and failure markers.					
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Figure 5: RQ4 frozen evaluation. Suites are not pooled into a single transfer score and cannot trigger model reselection.

6.5. RQ5: Which prespecified Behavior bases are associated with Utility?

RQ5 reuses the model family selected with B3 in RQ2. It does not reselect a model or a preferred feature group. Each group receives its own train-only first-trigger threshold within the same complete nested chain. The primary T0 comparison uses milestone rows only.

TBD The prespecified contrasts are B1–T0, B2–B1, B2+Motion–B2, B2+Maturity–B2, B3–(B2+Motion), and B3–(B2+Maturity). Function-level sign flips, hierarchical intervals, and coefficient-direction stability remain to be computed. For the six fixed BBOB validation functions, the six-contrast Holm family has minimum adjusted $p = 0.1875$ and cannot reject at 0.05; p values are transparency summaries, while the six function effects, finite-set mean, and conditional CI are primary.

TBD A Maturity increment can support only the predictive usefulness of these deterministic bases for the evaluated model/query. It cannot establish an observed search stage, causal mechanism, or universal analysis window.

Table 9

RQ5 six-group ablation using one frozen model family. The Motion and Maturity groups are siblings, permitting the two corresponding incremental contrasts.

Group	Inputs	Added in-formation	First-trigger Utility	AUROC/AP	Spearman/RMSE	Prespecified contrast
T0	1	FE ratio only	–	–	–	reference
B1	19	Core permutation-invariant Behavior	–	–	–	B1–T0
B2	25	Longitudinal set dynamics	–	–	–	B2–B1
B2+Motion	28	Three set-motion fields	–	–	–	B2+Motion–B
B2+Maturity	28	Three deterministic Maturity bases	–	–	–	B2+Maturity–E
B3	31	Motion and Maturity	–	–	–	two sibling incremental

TBD: add paired function effects, 95% CIs, six Holm-adjusted contrasts, call/terminal endpoints, and frozen BBOB-validation estimates. R

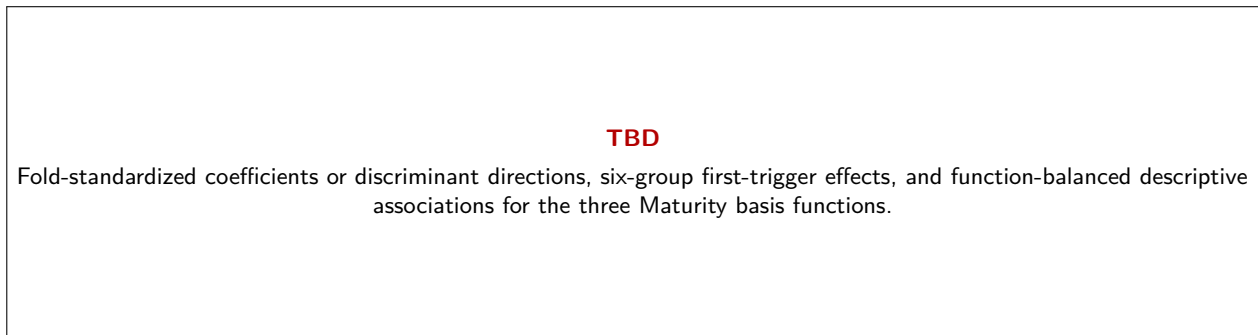


Figure 6: RQ5 Behavior interpretation. Coefficients and Maturity associations are predictive descriptions, not causal effects or evidence of a latent stage. No curve shape is prespecified.

Table 10

Robustness and diagnostic analyses that cannot replace a primary RQ result.

Analysis	Required output			Interpretation boundary
Query configurations	Complete for	parallel descriptor_cheap_invariant, pflacco_standard_invariant, and pflacco_broad_invariant	chains	Three evaluated representations only
Time weight	$\lambda_T \in \{0, 0.25, 0.5, 1, 2\}$ with terminal gap and runtime			Primary $\lambda_T = 1$ unchanged
Predictive query increment	OOF state-only	versus full-query	selected continuation-only gap	No sample best or acquisition cost
Operational query increment	Matched-trigger $I_q = U_q^{joint} - U_b$			Not a pure-information or causal effect
Action relations	selected_equals_default, selected_equals_prefix, handoff_required, transition type			No generic same/changed alias
Reservoir diagnostic	trajectory_query_reservoir_v1, zero extra FE			Not pooled with independent-LHS estimands

TBD: fill all prespecified estimates, intervals, coverage, and failures without choosing configurations after the outcomes are observed.

6.6. Prespecified robustness analyses and diagnostics

The primary manuscript cannot be completed until the two-stage replay loop is implemented: fit fold-specific selectors from one-pass action matrices, emit a fold-role replay plan, execute each selected decision-state future path three prespecified times for timing only, verify completed-replay endpoint consistency, retain any timing instability, and only then combine the Stage-A scientific endpoints with Stage-B median runtime to construct Utility. Current compute totals are boundary scenarios rather than a closed execution estimate because the nested replay multiplicity remains unimplemented.

7. Discussion

The present study separates two decisions that are usually collapsed in feature-based algorithm selection: whether to execute an evaluated fixed landscape query and, conditional on execution, which portfolio action to use. This formulation is methodological and does not by itself establish that the query should be skipped or executed in any empirical regime.

7.1. When is the evaluated query worth executing?

TBD The statewise utility distribution has not yet been estimated. Its eventual interpretation is conditional on the evaluated query, utility definition, effect direction, interval, and state coverage. The joint Query utility, full-budget Behavior-only utility, and their operational increment answer different questions and must be reported separately. The continuation-only predictive increment further excludes the sample best and acquisition cost; it cannot replace the operational comparison. None of these quantities can be extended from the fixed low-cost query to landscape analysis in general.

7.2. Information in algorithm-agnostic search behavior

TBD The RQ2 comparison between T0 and the behavior groups remains undetermined pending nested BBOB-train grouped-by-function OOF estimates and suite-specific evaluation. Decision utility, AUROC, Average Precision, Spearman correlation, and Ridge RMSE support different aspects of the comparison; disagreement among them precludes a single undifferentiated predictive claim.

7.3. End-to-end resource and performance trade-offs

TBD The practical resource–performance trade-off is undetermined. Its interpretation requires joint evidence from equal-total-FE final performance, decision-state-to-terminal future-path time, complete FE-zero-to-terminal policy wall-clock, measured query and inference time, query-call rate, and the cost of calls with non-positive observed utility. The conditional future-path time treats the shared prefix as a sunk cost; the full-run endpoint includes that prefix. They are not interchangeable. Fewer calls alone would not establish better resource-aware optimization.

7.4. Development sets and cross-benchmark scope

TBD Effects under the current protocol on previously inspected BBOB validation and CEC2017, and on prospective CEC2022 and engineering problems, remain unestimated and provide no current evidence of generalization. Any eventual interpretation is suite-specific; query-extraction failures, divergent effect directions, or incomplete coverage narrow its scope. The BBOB-validation estimand is specifically the equally weighted mean of its six fixed functions. It is a post-development internal finite-set estimate, not an independent confirmation or a sample from a function superpopulation. Function-resampling results, if shown, are composition sensitivities only.

7.5. Search maturity and behavior interpretation

TBD The RQ5 ablation, coefficient or discriminant-direction stability, and maturity–utility association remain

unestimated. Coefficients, discriminant directions, and correlations have a predictive rather than causal interpretation; a non-monotonic relationship would require support from its prespecified fit, interval, and stability analysis.

7.6. Limitations and validity boundaries

The primary estimand is query-specific: it concerns `descriptor_cheap_invariant`, a 14-dimensional custom low-cost descriptor query. Two response summaries that become constants under the common median/IQR preprocessing are intentionally excluded; this changes neither the evaluated sample nor its FE cost. The prespecified standard and broad `pflacco` queries can assess configuration robustness but cannot justify claims about untested landscape representations. Each problem/sample design uses one fixed problem-keyed LHS realization, so inference is conditional on those realizations rather than averaged over sampling randomness. Moreover, standard versus broad changes representation, sample size, sample realization, and budget together and cannot identify their separate contributions. The absence of a matched-acquisition state-only operational path likewise prevents descriptor-only attribution of I_q . The Selection Reference is a fixed downstream component rather than a claimed algorithm-selection contribution, and its selector regret must remain visible when interpreting query utility. The static VBS is selected at the problem level after averaging each algorithm over optimizer seeds; it never selects a different algorithm for every seed and is distinct from the statewise best observed action. The main Decision Model is trained offline with a BBOB grouped-by-function split; the historical grouping key does not denote a classic landscape family. The study does not evaluate online controller training as a primary design. Its primary protocol uses function–dimension–static–problem–run-balanced weights and divides run weight across eligible states. The estimator wiring is not yet implemented; nested first-trigger evaluation does not remove this fitting risk. Unweighted state-row fitting is retained only as a sensitivity analysis. All AAS and portfolio conclusions are conditional on the repository’s frozen DE, PSO, CMA-ES, and SHADE implementations, parameters, and population size 40.

TBD The empirical limitations remain to be quantified through trajectory and action-loss coverage, query-extraction failures, function-level dependence, population-transfer frequency, and sensitivity to the three prespecified query configurations and utility-cost settings.

8. Reproducibility

8.1. Software and experimental configuration

The study uses offline trajectory collection followed by supervised learning. The experimental specification fixes benchmark membership, dimensions, instances, optimizer seeds, portfolio composition, population size, FE budgets, and the dynamic decision-opportunity schedule. Reporting these elements, together with repetitions, variability, and feature cost, is particularly important for optimization representations [?].

The main implementation targets Python 3.12. The `pflacco` queries run in an isolated Python 3.11 environment with `pflacco==1.2.2`, `numpy==1.24.3`, `pandas==2.0.3`, `scipy==1.10.1`, and `scikit-learn==1.2.2`. The descriptor extractor receives only the previously evaluated LHS sample. It neither imports a benchmark nor calls the objective function, and a failed `pflacco` feature group is not replaced by a study-specific substitute.

8.2. Random streams

All scientific random streams are constructed from explicit integers through `numpy.random.SeedSequence`. Optimizer streams combine the configured optimizer seed with integer identifiers for the run and stochastic event. Query sampling additionally uses integer stream, function, instance, dimension, and sample-design codes. Population-transfer initialization receives explicit function, instance, and event codes. The Random Analysis repetitions combine a base seed with problem, dimension, optimizer seed, decision FE, and repetition indices. Experimental randomness therefore does not depend on strings or process-specific hash values.

One problem-keyed query realization is currently fixed for each problem and sample design and shared across optimizer seeds and states. Thus the primary release supports conditional inference for those realizations, not Monte Carlo averaging over query-sampling randomness.

8.3. Observation indexing and state alignment

An observation is uniquely indexed by

$$k_t = (\text{split}, p, \mathcal{F}(p), d, a_t, r, e_t), \quad (44)$$

where p uniquely identifies the problem instance, $\mathcal{F}(p)$ is its function-ID group, a_t is the prefix algorithm, r is the optimizer seed, and e_t is the exact integer FE count. This index aligns trajectories, behavior summaries, action outcomes, selection-reference predictions, utility labels, and Decision Model observations. Exact FE, rather than a rounded budget ratio, identifies a shared state; the corresponding elapsed-budget covariate is calculated as e_t/B . Nominal milestone ratios and realized native-window spans describe how the state was sampled but do not replace exact-FE alignment.

8.4. Data construction and evaluation dependencies

Complete native-state trajectories and complete-budget terminal outcomes define the common analysis population. Permutation-invariant behavior and the three fixed descriptor representations are evaluated at matched states, after which the Query-adjusted and Behavior-only full-budget four-action matrices are each executed once; the primary query also requires separate FE-zero four-algorithm outcomes for pre-run AAS. These prespecified Stage-A executions fix the scientific terminal gap, observed first hit, target-hit status, path completion, completion-conditioned endpoint success, planned FE, effective FE, and failure status,

fit fold-specific selectors, and produce an outer/inner/full-training replay plan. Only after the plan is frozen are the selected Skip, Query, and Behavior-only paths from their decision states to termination executed three prespecified times as Stage-B timing replays. Each query configuration has its own cross-fitted selection reference, utility labels, Decision Model, threshold, baseline comparison, and suite-specific evaluation. Only trajectories with both a complete native history and a terminal outcome enter the corresponding estimand. Query identity, sample design, FE charge, and descriptor definition jointly identify an analysis condition; in particular, the 5% and 10% acquisition conditions are analyzed separately and are never pooled.

At the present protocol stage, the replay planner can enumerate roles, but the offline decision-state-to-terminal runner has not been implemented and a fold-role-complete measured plan has not been materialized and checked. Both are formal run blockers; this manuscript does not imply that future-path timings or final Utility artifacts already exist. FE-zero AAS rows record the selected algorithm as the bookkeeping prefix, set selected-equals-prefix true, handoff-required false, and handoff type to fresh optimizer initialization; their default and no-query algorithm is the corresponding fold SBS. Outer-, inner-, and full-training Selector artifacts and their fold-role-to-model routing are also not yet persisted. A replay plan without that routing cannot measure the Selector inference actually used by each nested role.

8.5. Validity checks

The validity checks address state continuity, representation invariance, FE accounting, and information leakage. For each portfolio optimizer, a restored complete state is compared with uninterrupted native continuation at matched checkpoints, including optimizer-specific internal dynamics rather than only population values. Independent reordering of population members at every evaluated BBOB state tests that the 31 Decision Model inputs, three diagnostic variables, and native-window summaries are permutation invariant.

Query-level checks enforce the three descriptor definitions, the shared sample for the cheap and standard configurations, four non-duplicated actions, continuous remaining budget, query-specific FE accounting, the statewide action-loss transformation, and the utility and selector decompositions. Feature-group failures remain explicit; undefined individual descriptors are imputed only from BBOB-training observations. Model-level checks fit all preprocessing within the applicable training function-group fold, restrict the active candidates to LDA, Logistic Regression, and Ridge, and exclude BBOB validation and external suites from model fitting and threshold estimation. The three action relations in Eq. (38) are checked throughout selection, utility calculation, and policy evaluation. These procedures assess implementation validity; they add no observations to RQ1–RQ5. The primary fold-local weights equalize functions, fixed dimension strata, static problems, and optimizer runs

before dividing run weight among states. The current Selector and Decision estimators do not yet consume these weights; unit state-row fitting is only the planned sensitivity, so formal fitting is blocked until both estimator paths are wired consistently.

8.6. Resource measurement

FE accounting is exact at the state level:

$$FE_{\text{skip,opt}} = B - e_t, \quad (45)$$

$$FE_{q,\text{opt}} = B - e_t - FE_q. \quad (46)$$

Behavior extraction reuses evaluated optimizer states and makes no additional objective-function calls. Query time is the sum of sample-evaluation and feature-computation time. Selection latency includes state preparation, model inference, and action choice. Prediction time is measured separately for the T0 and B3 controllers.

The Utility time T_k is the censored median of three real executions from the common decision state to termination for $k \in \{\text{skip}, \text{query}, \text{behavior-only}\}$. It excludes the already executed common prefix; the Query path includes sampling, descriptor computation, selection, any population transfer, and continuation after that state. Policy reporting separately measures wall-clock from FE zero to termination, including native prefix updates, streaming Behavior extraction, every Decision inference, and any triggered path. The former is a state-conditional decision cost and the latter a complete policy-run endpoint. Field names and tables keep them separate.

Stage B determines only these wall-clock measurements. Every repetition records its order, component and complete-path time, status, effective FE, timeout, completion, and failure fields. Scientific gap, observed hit, path completion, endpoint success, planned FE, effective FE, and failure remain fixed to the prespecified Stage-A run. When the Stage-A path completed, every completed replay must reproduce its gap, observed hit, and first hit. When Stage A did not complete, completed replays are checked only against one another and never replace the failed scientific endpoint. Planned FE must preserve path identity; replay effective FE can differ and remains repetition-level timing metadata. A completed replay's censored time equals its raw observed time; a timed-out or failed replay's censored time is the larger of its raw time and role-specific timeout. The Utility time is the median of all three censored values, while the raw observed median is diagnostic only. Path identity, consistency among completed replays, and Stage-A-to-completed-replay consistency are stored separately. Within-Stage-B status instability and Stage-A/Stage-B completion instability are also stored separately. A legacy failure-worst-case field, if retained, is a strict alias of the censored value. No repetition is selectively rerun, deleted, or replaced after its time or status is observed. The same separation applies to FE-zero policy runs.

All resource measurements use one fixed hardware and software environment. Processor, operating system, Python environments, thread count, batch size, cache state, and the peak-memory procedure accompany the estimates. Timings

are decomposed into behavior extraction, T0 and B3 inference, query sampling, descriptor computation, downstream selection, decision-state future-path time, and FE-zero-to-terminal policy wall time. **TBD**

8.7. Recomputation of estimands

The state-level release retains the quantities needed to recompute every term in Eq. (12) and Eq. (22), including the observed losses, normalization and cost terms, selected action, and three action relations. Model summaries are reconstructed from training function-group fold assignments, nested and full-training OOF scores, thresholds, and suite-evaluation predictions. Policy summaries are reconstructed from the trigger state, query decision, selected action, transition mode, Stage-A FE and terminal loss, Stage-B timing components and repetition statuses, and the timing-instability indicator. The release also identifies outer, inner, and full-training replay roles so that selected-path multiplicity and every timing observation can be reconstructed; an action-matrix row alone is not treated as a completed Utility observation.

BBOB-validation interval reconstruction keeps all six fixed functions, all fixed dimensions, and all static problems for instances 1–3 in every primary bootstrap replicate, resampling paired optimizer seeds only within each fixed static problem. Function-resampling uses a distinct sensitivity label and is never substituted for the primary conditional interval.

ERT reconstruction uses a dedicated paired ratio bootstrap that recomputes FE numerators and target-hit counts within every function–dimension cell in every replicate. One-policy zero-hit cells are retained as signed infinities and two-policy zero-hit cells as undefined, with the interval marked unestablished. Although the helper exists, it is not yet connected to the suite-level attempted-coverage, failure-sensitivity, and reporting consumer; complete-pair ERT ratios alone cannot establish a suite conclusion.

The numerical margins are interpreted as study-specific operational tolerances. Ordinary and family-wise simultaneous intervals are descriptive sensitivity summaries of their position relative to those margins; the first paper makes no confirmatory equivalence claim. Previously inspected BBOB validation and CEC2017, prospective CEC2022, and each engineering suite are analyzed separately with every train-derived component fixed. These provisions define reproducibility and do not themselves constitute evidence of cross-function or cross-benchmark transfer.

9. Conclusion

This work formulates analysis selection as a decision that precedes the execution of a prespecified fixed landscape query. The proposed offline workflow links full-state optimizer trajectories, permutation-invariant behavior states, statewise observed action losses, a fixed Selection Reference, and a supervised Decision Model while preserving equal total FE budgets. It distinguishes the joint Query path utility from both the different-budget Behavior-only

operational increment and the same-budget continuation-only predictive increment of query descriptors. It also distinguishes decision-state future path time from FE-zero-to-terminal policy wall-clock. These design elements define the evaluation; they do not substitute for the empirical evidence required by RQ1–RQ5.

TBD Whether the primary query has non-positive utility in a statistically supported subset of states remains undetermined; the required effect size, interval, and query-specific scope are not yet available.

TBD Whether algorithm-agnostic behavior predicts query utility beyond FE ratio alone remains undetermined pending nested grouped-by-function OOF evidence and suite-specific evaluation.

TBD Effects on calls with non-positive utility, mean decision utility, measured runtime, and equal-budget final optimization performance relative to the prespecified policies remain undetermined.

TBD No conclusion under the current protocol is currently supported for the fixed six BBOB-validation functions or any external suite. Any BBOB-validation estimate remains a previously inspected internal finite-set mean and not an independent confirmation or function-superpopulation claim; CEC2017 is likewise a previously inspected external development set. CEC2022 and engineering-problem generalization remain unverified pending prospective estimates and coverage analyses after their configurations and endpoints are fixed.

TBD Behavior-group and search-maturity associations remain undetermined and, when estimated, have a descriptive interpretation conditional on the prespecified representations.

The contribution is limited to the prespecified 14-, 37-, and 52-descriptor query configurations. Consistent evidence from these three configurations would support configuration robustness, not a universal conclusion about all landscape analysis methods or representations.